Three-way convergence analysis — final report

Recurrent CNVs (Sacks et al.) × exome height genes (Kosmicki et al.) × height-decreasing effect bias

PGC joint submission · generated 2026-09-20 · R 4.6.1 · no pathway statistic recomputed

1. Summary

Across an analysis set of **102 pathways**, GO:0008284 — Positive Regulation Of Cell Population Proliferation is the **only pathway significant in all three tests of the asymmetric analysis**: enrichment in asymmetric-CNV genes (B), enrichment of the pathway within the 207 Kosmicki height genes (C), and bias toward height-decreasing effects in the pathway's Kosmicki genes relative to all 207 Kosmicki genes (D).

It is **Bonferroni significant in test C** ($P = 1.05\times10^{-4}$ against $0.05/102 = 4.9\times10^{-4}$; BH q = 0.0045), and ranks **first by P in both the asymmetric-CNV test and the effect-direction test**.

The same analysis run with **mirror CNVs** yields **no convergent pathway**: none of the 102 is significant in all three of A, C and D.

Test	GO:0008284	P	q (BH)	P<0.05	Bonferroni
A — Pathway enrichment in mirror CNVs	3 genes, OR 0.44	0.94	1.00	3	0
B — Pathway enrichment in asymmetric CNVs	9/10 universe genes, OR 7.04	0.031	1.00	2	0
C — Pathway enrichment in Kosmicki height genes	17/176 genes, OR 3.11	**1.05×10^{-4}**	**0.0045**	22	3
D — Negative-effect bias vs all 207 Kosmicki genes	16 neg / 1 pos, OR 8.70	0.0090	0.70	3	0
Convergent (asymmetric: B, C and D)	—	—	—	**1**	—
Convergent (mirror: A, C and D)	—	—	—	**0**	—

2. Pathway set

Pathways with statistics (5,712)

Two gene-set libraries were read as GMT files. Gene symbols were upper-cased and de-duplicated within each term, and a term was retained only if it contained between **5 and 500** unique symbols (`min_set_size = 5` , `max_set_size = 500` in `kosmicki_pathway_cnv_pipeline.R` , applied when the GMT is parsed).

Library	Terms in GMT	Dropped (<5 or >500 genes)	Retained	Gene universe
GO Biological Process 2023	5,407	14	5,393	14,212
KEGG 2021 Human	320	1	319	8,077

Total with statistics	5,727	15	**5,712**	—

Verified by re-parsing the original `GO_Biological_Process_2023.gmt` and `KEGG_2021_Human.gmt` and re-applying the size filter, which reproduces 5,393 and 319 exactly.

Analysis set (102)

The CNV gene universe is small — **215 mapped genes for GO:BP** (124 asymmetric, 104 mirror) and 103 for KEGG (61 asymmetric, 49 mirror) — so a pathway with few genes in that universe cannot reach P < 0.05 in a CNV test however extreme its genes are. Pathways were therefore restricted to those carrying enough CNV genes for a CNV test to be capable of nominal significance. The threshold is **derived, not chosen**: for each pathway the best-case attainable P — all of its CNV-universe genes belonging to the target class — is computed from the hypergeometric distribution, and the pathway is admitted if that is below 0.05 for either CNV class.

Genes in CNV universe	1	2	3	4	5	6	7	8	10
Best-case attainable P, asymmetric (GO:BP)	0.58	0.33	0.19	0.11	0.062	**0.035**	0.020	0.011	0.0035
Best-case attainable P, mirror (GO:BP)	0.48	0.23	0.11	**0.053**	0.025	0.012	0.0056	0.0026	0.00056

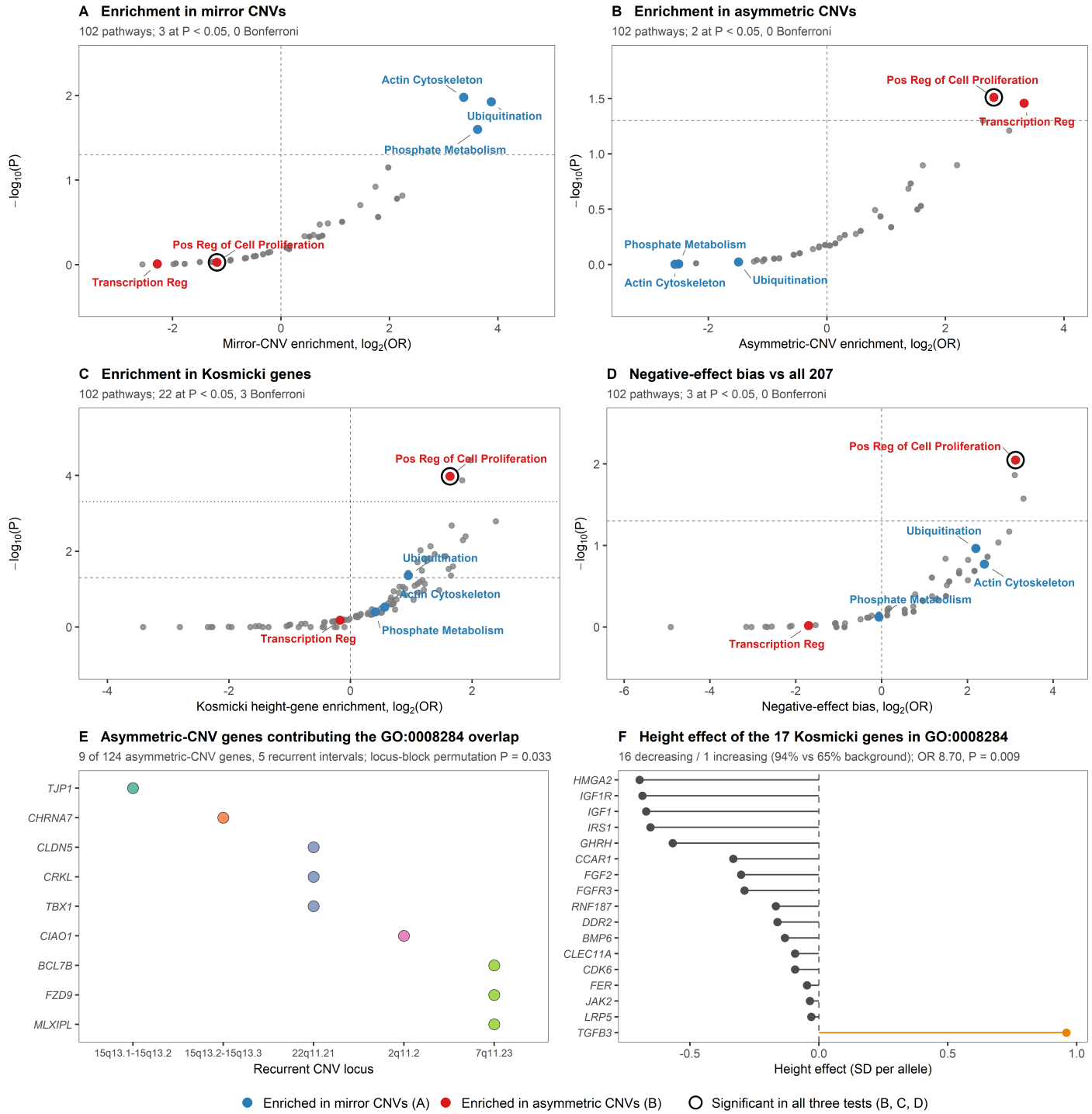
> **Analysis set = 102 pathways** (68 GO:BP + 34 KEGG): the **union** of the 48 pathways attainable for the asymmetric test (≥ 6 CNV-universe genes) and the 102 attainable for the mirror test (≥ 4). The asymmetric set is a strict subset of the mirror set, so the union is the mirror set. **This one set is used for all four tests** and fixes the multiple-testing burden throughout: Bonferroni = 0.05/102 = 4.9×10^{-4}, with BH FDR computed within it. Tests C and D do not involve CNVs, so they are computed once and shared by the two convergence analyses; only the CNV test differs between them.

GO:0008284 has 10 of its genes in the CNV universe, 9 of them asymmetric and 3 mirror.

3. Main figure

Three-way convergence on a single analysis set: pathway enrichment in mirror (A) and asymmetric (B) CNVs from Sacks et al., pathway enrichment in height genes from Kosmicki et al. (C), and enrichment of height-decreasing effect bias in the pathway's Kosmicki genes relative to all 207 Kosmicki genes (D), with the gene-level evidence behind the convergent pathway GO:0008284 (E, F)

Panels A-D plot the same 102 pathways: the union of those for which the asymmetric CNV test can reach P < 0.05 (>= 6 genes in the CNV gene universe) and those for which the mirror CNV test can (>= 4). That single set fixes the multiple-testing burden throughout, Bonferroni = 0.05/102 = 4.9e-04. In A-D colour marks the three pathways enriched in mirror CNVs (blue) and the two enriched in asymmetric CNVs (red); these groups are disjoint, and only these five carry labels, in the same colours in every panel so each can be followed across the tests. Significance within a panel is read off the threshold lines. The black ring marks the 1 pathway significant in all three tests of the asymmetric analysis (B, C and D): GO:0008284. Panels E and F are gene-level and use their own palettes.



Dashed lines mark OR = 1 and nominal P = 0.05; dotted line marks the Bonferroni threshold where it falls within the plotted range. Neither CNV test shows a screen-level excess (asymmetric 2 hits, mirror 3; chance expects 2.4 for the asymmetric test, which only 48 of the 102 pathways can reach). The contrast between the two CNV classes is a specificity check rather than a formal test. In E, genes within one recurrent locus travel on the same CNV and are not independent, which is why the locus-block permutation, not the gene-level Fisher test, is the statistic of record for the CNV layer.

Figure 1. (A) pathway enrichment in mirror CNVs; **(B)** in asymmetric CNVs; **(C)** pathway enrichment in the 207 Kosmicki height genes; **(D)** negative-effect bias in the pathway's Kosmicki genes relative to all 207. Panels A–D plot the same 102 pathways. Colour there marks

membership of the two CNV-enriched groups rather than significance in the panel being viewed: **blue**, the three pathways enriched in mirror CNVs (A); **red**, the two enriched in asymmetric CNVs (B); grey, all others. The two groups are disjoint, and those five pathways, and only those five, carry text labels in the same colours in every panel, so each can be followed across the tests. Significance within a panel is read off the threshold lines: dashed, OR = 1 and P = 0.05; dotted, the Bonferroni threshold (4.9×10^{-4}) where it falls within the plotted range. The black ring marks the one pathway significant in all three tests of the asymmetric analysis (B, C and D), GO:0008284. **(E)** The 9 asymmetric-CNV genes contributing the GO:0008284 overlap, grouped by recurrent locus — genes within one locus travel on the same CNV and are not independent, which is why the locus-block permutation (P = 0.033) rather than the gene-level Fisher test is the statistic of record for the CNV layer. **(F)** Height effect of the 17 Kosmicki genes in GO:0008284, ascending: 16 height-decreasing (grey), 1 height-increasing (*TGFB3*, +0.96 SD, orange), against a 64.8% background rate across all 207. Panels E and F are gene-level and use their own palettes, deliberately avoiding the blue/red of A–D.

Pathways at each intersection

Pathways in each exclusive intersection of the three tests, per analysis. Regions not listed are empty.

Analysis	Intersection	Pathway number	Pathway	Database
Asymmetric	**B|C|D**	1	**Positive Regulation Of Cell Population Proliferation (GO:0008284)**	GO:BP
	B|C	—	*none*	—
	C|D	1	PI3K-Akt signaling pathway	**KEGG**
	C|D	2	Positive Regulation Of MAPK Cascade (GO:0043410)	GO:BP
Mirror	**A|C|D**	—	***none***	—
	A|C	1	Protein Ubiquitination (GO:0016567)	GO:BP
	C|D	1	Positive Regulation Of Cell Population Proliferation (GO:0008284)	GO:BP
	C|D	2	PI3K-Akt signaling pathway	**KEGG**
	C|D	3	Positive Regulation Of MAPK Cascade (GO:0043410)	GO:BP

B|D and A|D are empty and omitted. C|D differs between the analyses (2 vs 3) because in the asymmetric analysis one of the three C∩D pathways also clears B and moves to the three-way intersection; C∩D is 3 in both.

> **No gene in any of these pathways is both a Kosmicki height gene and a Sacks CNV gene.** The overlap in every case is between two disjoint gene sets within the same pathway, so the three tests are not being driven by the same genes.

4. Rank of the convergent pathway

Where GO:0008284 sits within the 102-pathway analysis set in each test:

Test	P	Rank by P	OR	Rank by OR
A — mirror-CNV enrichment	0.94	88 of 102	0.44	83 of 102
B — asymmetric-CNV enrichment	0.031	**1 of 102**	7.04	3 of 102

C — Kosmicki enrichment	1.05×10^{-4}	2 of 102	3.11	9 of 102
D — negative-effect bias	0.0090	**1 of 102**	8.70	2 of 102

GO:0008284 is the single most significant pathway in both the asymmetric-CNV test and the effect-direction test, second in the Kosmicki test, and sits in the lower half of the mirror-CNV test.

5. Suggested wording for the manuscript

We restricted the analysis to pathways carrying enough genes in the recurrent-CNV gene universe for a CNV test to be capable of nominal significance (≥ 6 of the 215 mapped genes for the asymmetric test, ≥ 4 for the mirror test; 102 pathways in union), and used this single set for all tests, so that the multiple-testing burden is set once (Bonferroni $P < 4.9\times10^{-4}$).

Of these 102 pathways, Positive Regulation Of Cell Population Proliferation (GO:0008284) was the only one significant in all three tests of the asymmetric analysis: enrichment among genes in asymmetric CNVs (OR = 7.04, P = 0.031), enrichment within the 207 exome-wide significant height genes (17 of 176 mapped genes, OR = 3.11, $P = 1.05\times10^{-4}$, Bonferroni significant, FDR = 0.0045), and a bias toward height-decreasing effects among those genes relative to all 207 (16 of 17 negative, OR = 8.70, P = 0.009). It ranked first by P in both the asymmetric-CNV test and the effect-direction test.

The nine contributing genes fall in four recurrent loci (15q13, 22q11.21, 2q11.2, 7q11.23); a locus-block permutation preserving the correlation among genes on the same CNV confirmed the excess (P = 0.033). The enrichment was unchanged after excluding the height genes themselves (OR = 7.03, P = 0.031), indicating that the CNV and exome signals are independent, and was absent in mirror CNVs (OR = 0.44, P = 0.94).

Repeating the analysis with mirror CNVs in place of asymmetric CNVs identified no pathway satisfying all three criteria. Neither CNV test showed an excess of enriched pathways in aggregate (2 asymmetric and 3 mirror pathways at P < 0.05), so we do not interpret either CNV screen as showing pathway enrichment across the CNV set; the evidence for GO:0008284 rests on its convergence across the three tests, on the Bonferroni-significant enrichment in the height gene set, and on the locus-block permutation.

6. Supporting and sensitivity analyses

Analysis	Estimate	P	Interpretation
Asymmetric-CNV enrichment	OR 7.04 (9 of 10 universe genes)	0.031	Nominal
Locus-block permutation	Δfraction 0.044 (null 0.003 ± 0.024)	**0.033**	**Statistic of record for the CNV layer**
Mirror-CNV enrichment	OR 0.44 (3 genes)	0.94	Specific to asymmetric loci
Asymmetric vs mirror, direct contrast	OR 6.06	0.059	Same direction, borderline
Asymmetric enrichment excluding Kosmicki genes	OR 7.03	0.031	CNV and exome layers independent
Kosmicki enrichment	OR 3.11	**1.05×10^{-4}**	**Bonferroni significant; q = 0.0045**
Negative-effect bias, two-sided Fisher	16/1	0.013	Robust to test choice

Negative-effect bias, Mann–Whitney	—	0.0090	Robust to test choice
Independent-gene sensitivity	16/1, OR 8.71	0.0090	Unchanged
Redundancy pruning	cluster size 1	retained	Not a duplicate of another GO term

The **17 Kosmicki genes**, by effect size: *HMGA2*, *IGF1R*, *IGF1*, *IRS1*, *GHRH*, *CCAR1*, *FGF2*, *FGFR3*, *RNF187*, *DDR2*, *BMP6*, *CLEC11A*, *CDK6*, *FER*, *JAK2*, *LRP5* (all height-decreasing), and *TGFB3* (height-increasing). Dominated by the GH–IGF growth axis and FGF signalling. The **9 asymmetric-CNV genes**: *TJP1* (15q13.1–13.2), *CHRNA7* (15q13.2–13.3), *CLDN5*, *CRKL*, *TBX1* (22q11.21), *CIAO1* (2q11.2), *BCL7B*, *FZD9*, *MLXIPL* (7q11.23).

7. Caveats

> - Neither CNV test shows an excess of enriched pathways in aggregate: 2 asymmetric and 3 mirror pathways at P < 0.05, and none survives correction in either. The CNV screens should not be presented as evidence of pathway enrichment in their own right.
> - Only the Kosmicki enrichment (test C) is Bonferroni and FDR significant. Tests A, B and D are nominal.
> - The contrast between the asymmetric and mirror analyses is a specificity check, not a formal test: one convergent pathway versus none is not by itself a significant difference, and the direct asymmetric-vs-mirror contrast for GO:0008284 is borderline (OR = 6.06, one-sided P = 0.059).
> - Tests C and D are not independent: both are computed on the Kosmicki gene set.
> - *CHRNA7* appears in both an asymmetric (15q13.1–13.2) and a mirror (15q13.3) interval in the annotation table; it was not relabelled.

Data-quality note. The `*_overlap_genes` string columns in the master results table are unreliable — 1,286 asymmetric and 967 mirror rows carry a gene string whose length disagrees with that row's own `overlap_n`, and many are blank. Gene-level results here are therefore reconstructed from the source GMT files plus the two annotated gene tables, and validated against the master `overlap_n` columns (0 mismatches across all 5,712 pathways for the Kosmicki, asymmetric and mirror sets). `make_intersection_table.R` re-runs that validation and stops on any mismatch.

8. Files and reproduction

From `Final/`, in order:

```
Rscript run_three_way_convergence.R      # analysis set, four tests, ranks, logs
Rscript make_main_figure.R               # Figure 1
Rscript make_intersection_table.R        # pathways at each intersection
```

File (in `results/`)	Contents
`00_analysis_log.txt`	Full analysis log: set derivation, the four tests, convergence, ranks
`Figure_1_three_way_convergence.png/.pdf`	**Figure 1**, 6 panels (A-D pathway level, E-F gene level), 400 dpi

`analysis_set_102_pathways.csv`	The 102 pathways: all four tests, q-values, Bonferroni and convergence flags
`analysis_summary.csv`	Headline counts and thresholds
`analysis_set_attainability.csv`	Best-case attainable P by genes-in-universe, for both CNV classes
`convergent_pathways.csv`	Pathways significant in all three tests of either analysis
`convergent_pathway_ranks.csv`	Rank of GO:0008284 by P and by OR in each test
`intersection_pathways.csv`	Pathways at each intersection, with gene counts and shared genes
`testA_mirror / testB_asymmetric / testC_kosmicki / testD_bias_significant.csv`	Pathways significant in each test

Inputs: `kosmicki_pathway_cnv_master_results.csv`, `kosmicki_207_genes_annotated.csv`, `cnv_genes_annotated.csv`, and the two GMT files in `../Old/original_pipeline_review/original_bundle/`.